

Lab Safety Symbols and Hazard Signs, Meanings

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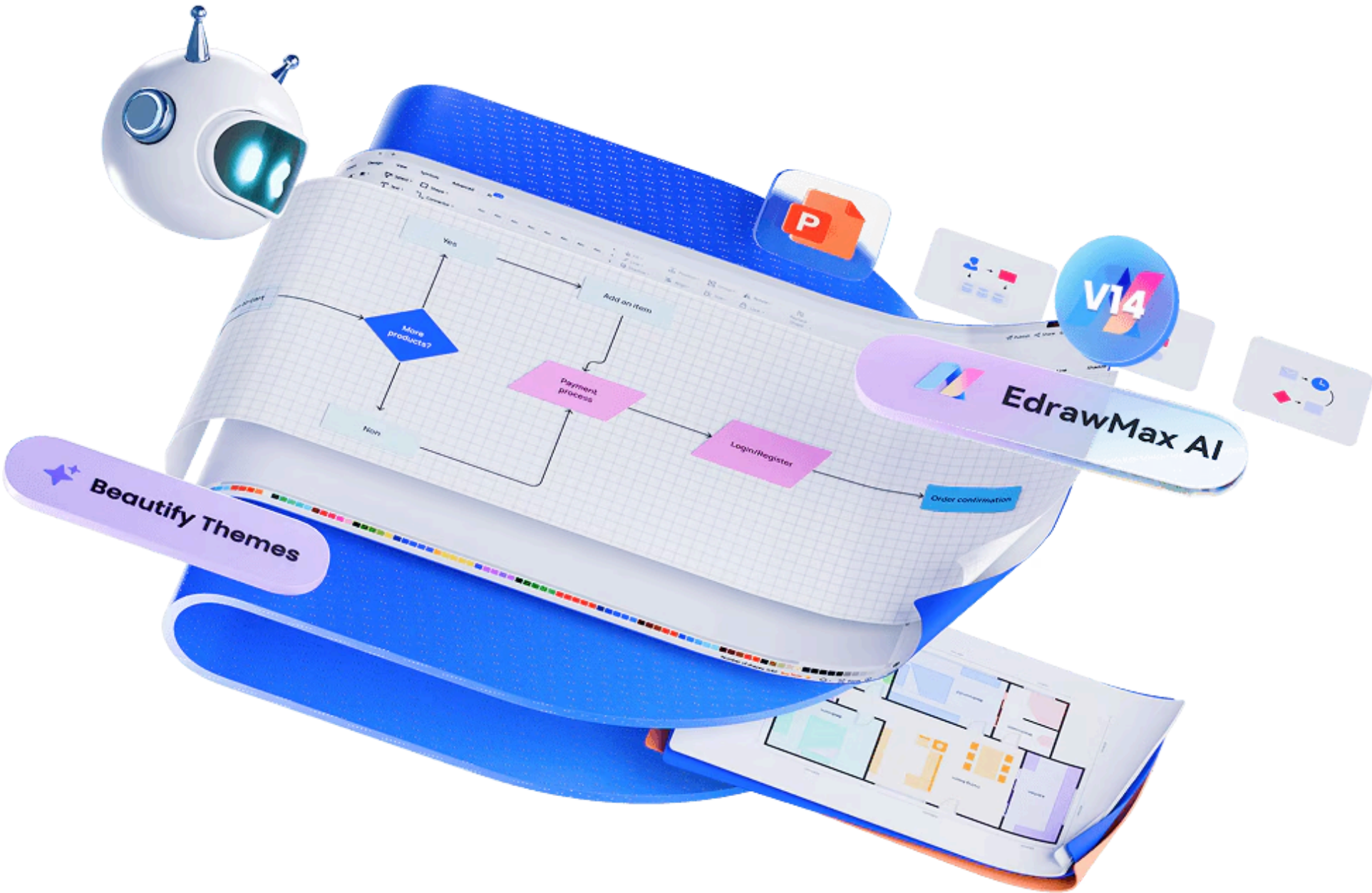
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How to interpret the signs: first identify whether the sign communicates a chemical hazard, required action, prohibition, emergency equipment, radiation/biohazard risk, or general warning. A symbol is only a visual cue; laboratory procedures, labels, Safety Data Sheets (SDS), and local rules remain authoritative for a specific substance or task.



Lab safety symbols are extremely important for keeping people safe in the lab and protecting lab equipment and instruments. There are two main aspects of lab safety that we need to consider.

The first aspect is understanding the potential dangers and risks in the lab. For example, some labs may have radioactive substances, dangerous chemicals, fire hazards, and sharp instruments. Lab safety symbols, like warning signs, are used to alert people to these hazards. It's crucial that everyone working in the lab understands these symbols, just like it's important for drivers to understand traffic signs on the road.

The second aspect is about taking precautions and knowing what to do in case of an accident. This includes wearing protective gear like gloves and safety goggles, having first aid kits available, and receiving first aid training. These measures are in place to ensure that lab workers are protected and can respond effectively if something goes wrong.



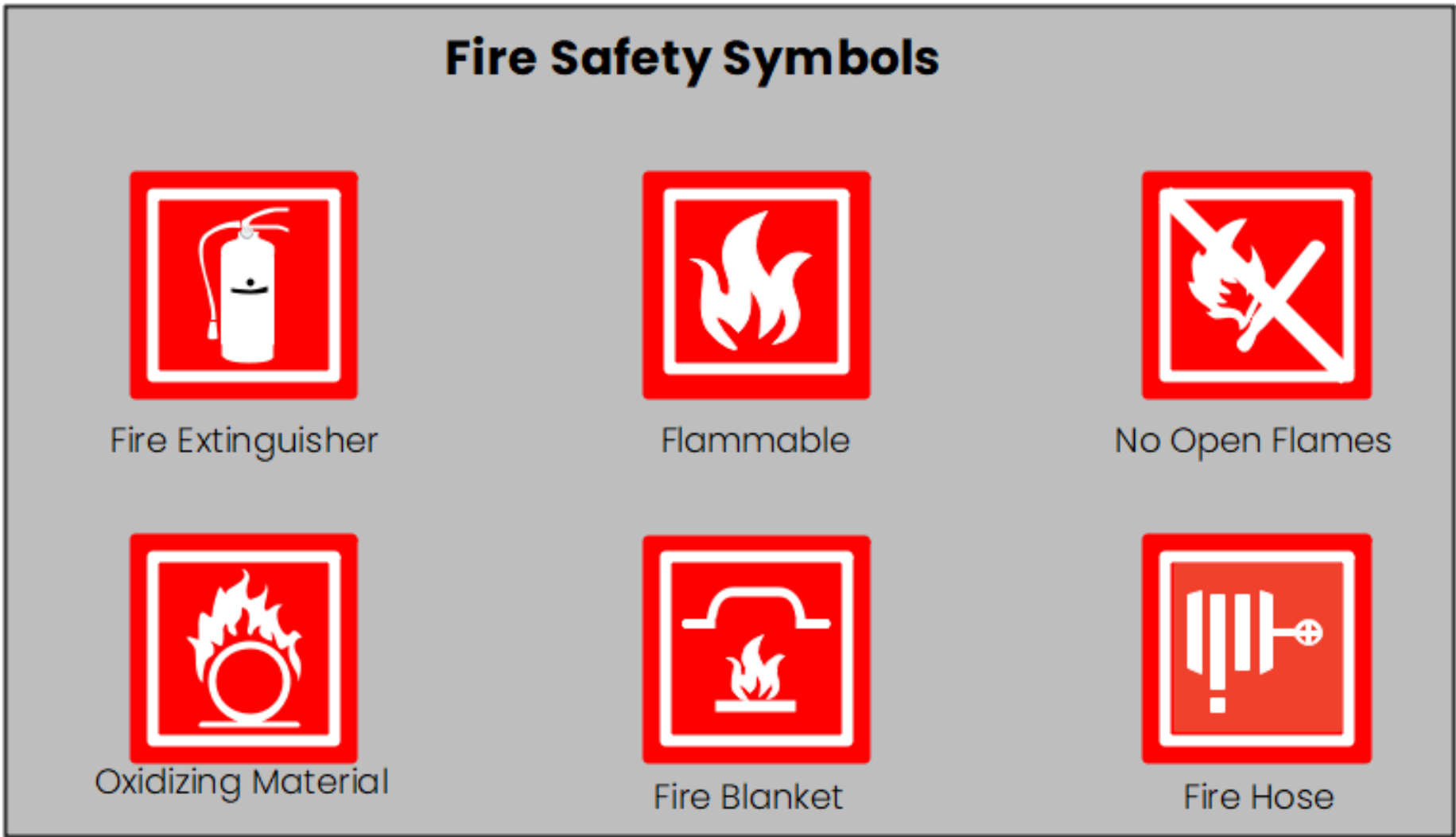
Lab Safety Symbols



2. Lab Safety Symbols Explained

2.1 Fire Safety Symbols

Fire safety symbols support people's safety in case of a fire accident. Also, the firefighters and emergency teams can find their way and devise a better fire safety plan. These symbols include fire extinguishers, fire hoses, fire blankets, etc. let us see the details.



Fire Extinguisher: It is used to mention where the fire extinguisher is placed. It may also include the audit details of the extinguisher.

Flammable: It is highly used and critical in lab safety symbols because there are many flammable substances in labs. It is important to label flammable things for safety.

No Open Flames: It is used where any substances are placed that can catch fire easily. These substances include fuels, volatile liquids, and so on.

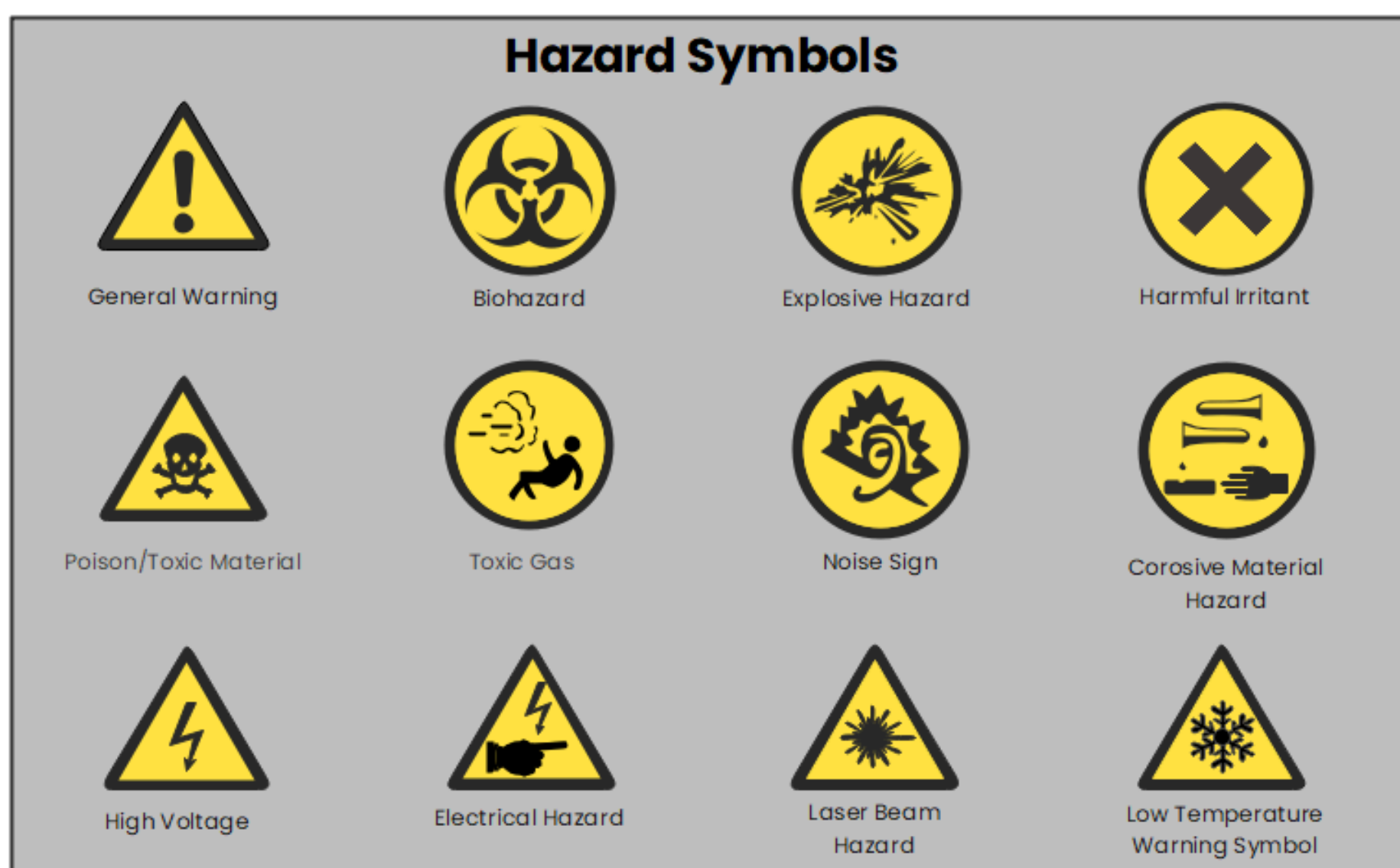
Fire Blanket: It is a piece of safety equipment that helps in controlling fire.

Fire Hose: It marks the place of the fire hose in the lab.

Oxidizing Material: Oxidizing materials can give oxygen to chemical substrates and flammable substances. Lab team must place oxidizing materials and flammable substances at a distance. Proper safety equipment is important while dealing with oxidizing materials.

2.2 Hazard Symbols

Hazard symbols are the lab safety symbols that show the presence of a material or equipment that is harmful. The most frequently used symbols are;



Hazard Symbols



General Warning: The general warning symbol indicates potential risks and reminds the person dealing with it to use PPE and other safety precautions.

Biohazard: The "biohazard" symbol shows that this particular place deals with biological substances—for example, body secretions, bacteria, and samples.

Explosive Hazard: The "explosive material" warning indicates that materials that can explode are present here. These materials include aerosols and compressed gases and materials that may explode because of exposure to high heat or flame and mixing with other substances.

Harmful Irritant: Substances that can irritate the skin, eye or respiratory tract irritant, or narcotic.

Poison/Toxic Material: Examples of poison or toxic materials include ammonia, bleach, and battery acid. The extent of danger depends on the concentration of material, duration of exposure, and environmental conditions.

Toxic Gas: Inhaling these gases is a health hazard.

Noise Sign: It is a sign for quiet areas where noise must be controlled.



High Voltage: It is a sign for high voltage current where caution must be taken.

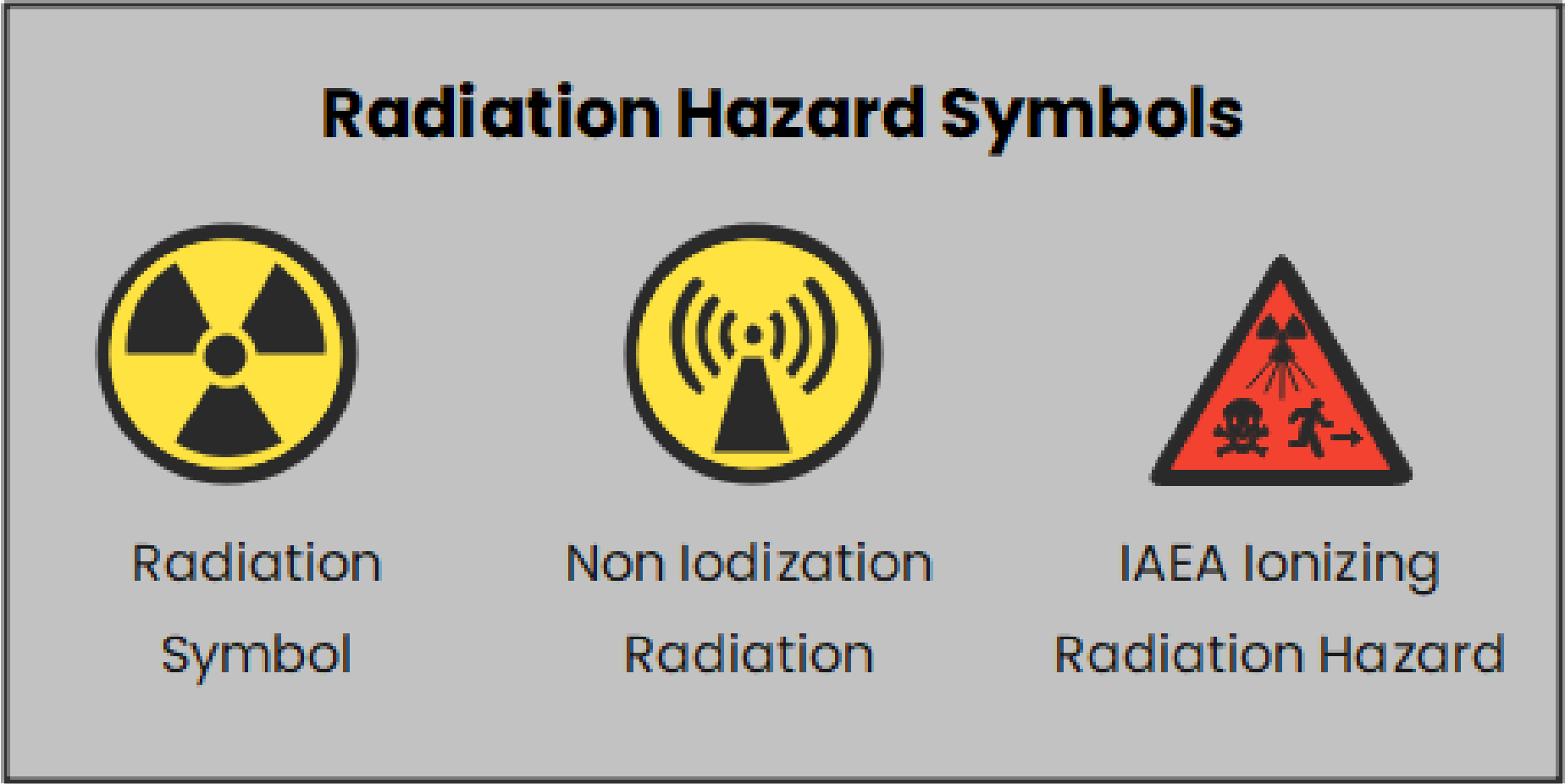
Electrical Hazard: These signs indicate electrical hazards like concealed high voltage wires and equipment or buried cables etc.

Laser Beam Hazard: High-intensity focused laser beams can be dangerous for human eyes and skin.

Low-Temperature Warning Symbol: These symbols are for low temperature and cryogenic hazards. These substances are below freezing point and stored with chemicals like liquid nitrogen.

2.3 Radiation Hazard Symbols

Radiation Hazard Symbols warn about the radiations that may break chemical bonds. There are several potential health risks associated with exposure to such substances.



Radiation Hazard Symbols



Ionizing Radiation Hazard: Ionizing radiation hazards include skin burns or acute radiation syndrome when radiations go beyond a certain level.

Non-Ionizing Radiation Hazard: Non-ionizing radiation hazards include ultraviolet, visible light, infrared, microwave, radiofrequency, and extremely low frequency (ELF).

IAEA Ionizing Radiation Hazard: It is usually placed on the device housing the radiation source to warn people not to take the device apart or get any closer.

2.4 First Aid Symbols

First aid symbols include the critical lab safety symbols that help quick and efficient first aid services in case of accidents or emergencies.

First Aid Station: It represents a station where first aid is available before proper medical care is provided.

AED or Defibrillator: It represents a device that is used to help those experiencing sudden cardiac arrest.

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2.5 Physical Safety Symbols

Physical safety symbols provide lab staff access to the equipment they can use to keep themselves safe from hazards.



Physical Safety Symbols



- Breathing Masks:** Breathing masks help the staff filter the airborne hazardous materials from the air and inhale clean oxygen.
- Eyewash Station:** Eyewash stations provide washing and first aid in case of any splash of hazardous liquid in the eyes or any other eye injury.
- Food and Drink Prohibited:** Food and drink prohibited signs are placed to provide safety in two domains. Exposed food can get contaminated with chemicals and other hazardous materials in the lab. Also, this lab safety symbol is helpful where the delicate sample may get infected with food particles.
- Gloves Required:** Gloves required for lab safety symbols are used in settings with potential risks of chemical burns, corrosion, allergic reactions, etc. It mentions that the gloves must be used before handling this stuff.
- Safety Glasses:** Potential eye safety hazards are present so use safety glasses.
- Protective Clothing:** Different types of Protective clothing may be required for radioactive hazards, bacteria and virus exposure, extreme temperatures, chemical exposures, and so on.
- Boots Required:** This lab safety symbol indicates that you need to use anti-slip, waterproof, and anti-static boots before entering this area.
- No Pacemakers or Metallic Implants:** This sign indicates that this area may contain strong magnetic fields, and any individuals with pacemakers or metallic implant devices should avoid working here.

3. Tips for Using Lab Safety Symbols

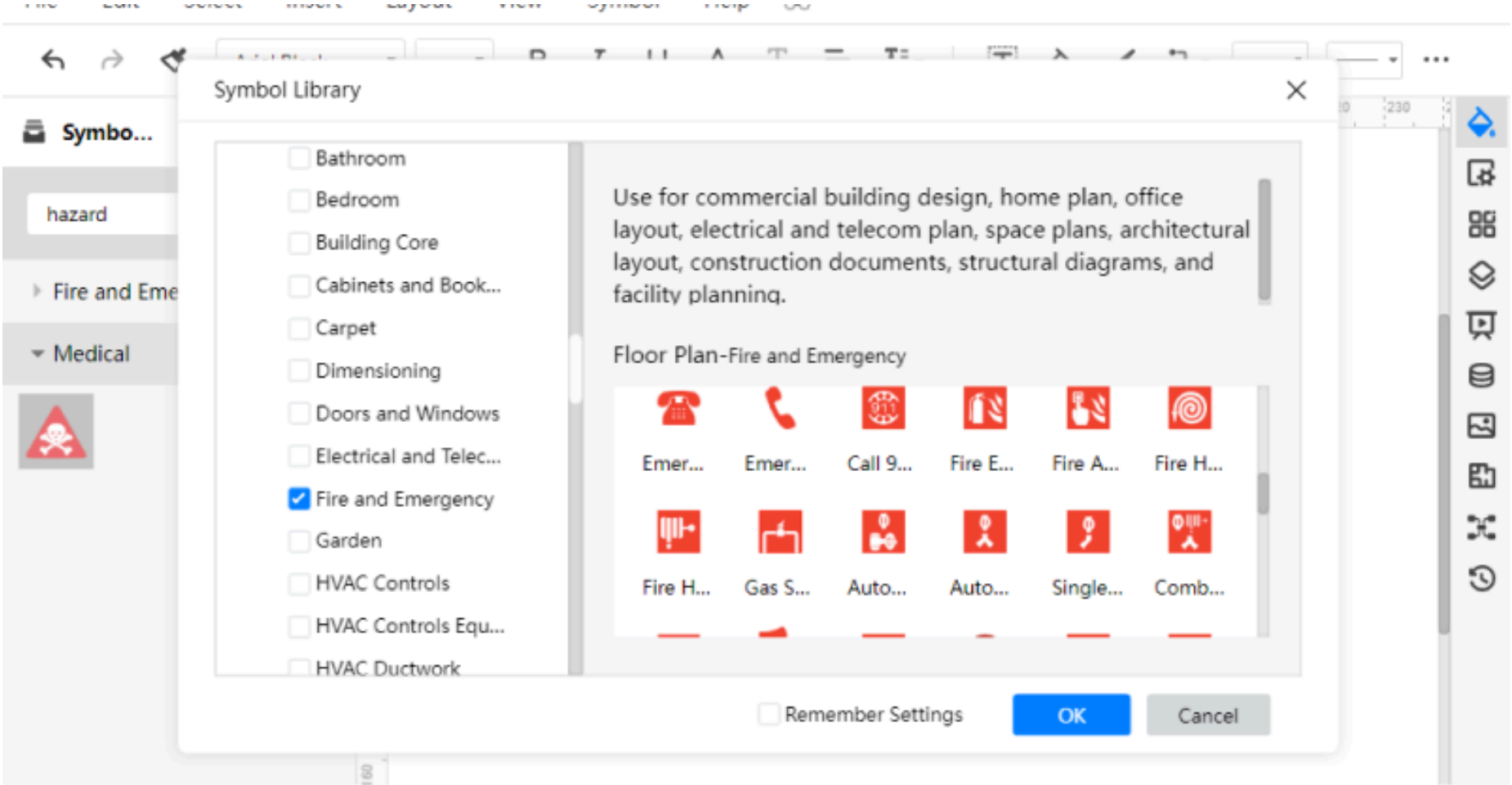
EdrawMax Online is an easy-to-use yet professional and comprehensive support for 280+ different diagrams, including lab safety symbols drawings. The learning curve will not bother you as it takes only a few clicks to get the task done. Here are some tips.

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1. In the new drawing window, click the symbols library button on the sidebar and navigate to Science items. Here, you will find different categories for science, click on the laboratory equipment tab and go ahead with your creativity.



2. If you think the symbol for lab safety is not available in the library, you can also import it. It is very simple, and you can use SVG, VSSX, and image formats. It is very easy to use the import interface. In the sidebar, click on the import icon. A pop-up window appears; go to the tab according to the symbol file format you want to import.

4. Conclusion

Lab safety symbols are not like other symbols; correctness and error-free drawing is critical because it saves lives and prevents accidents. The main categories of these symbols are fire safety, hazard, radiation hazard, first aid, and physical safety symbols. EdrawMax Online has a comprehensive library of lab safety symbols, [science graphic organizer examples](#) to help teachers, students, scientists, lab assistants, and other professionals to create diagrams with authentic safety symbols for the lab.

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